SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA5612

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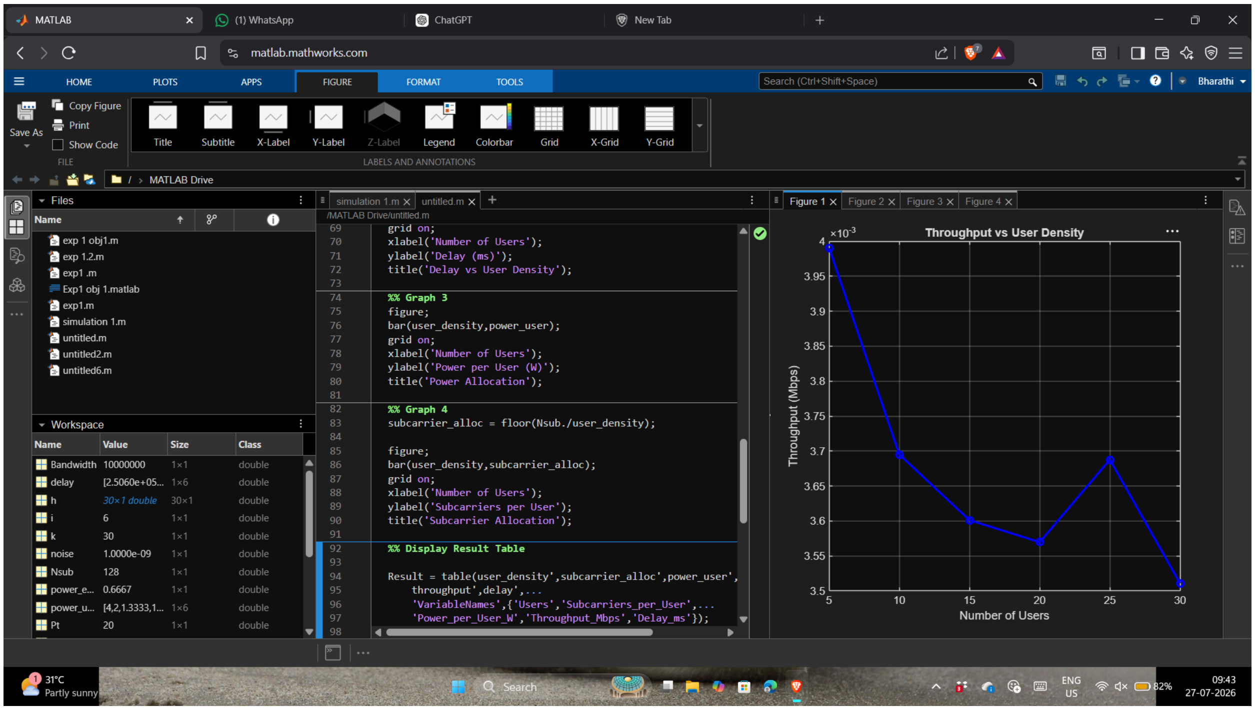
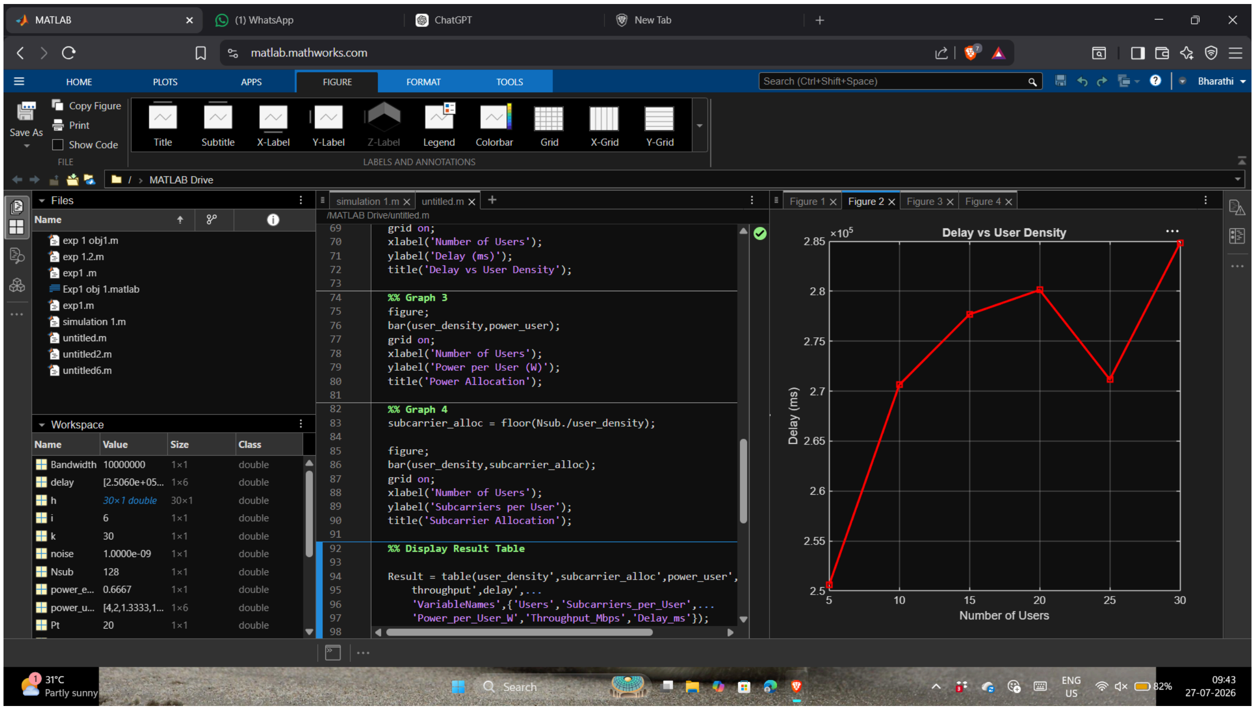
19. Simulation of OFDMA resource allocation under user density, subcarriers, power allocation, throughput, and delay (BL4)

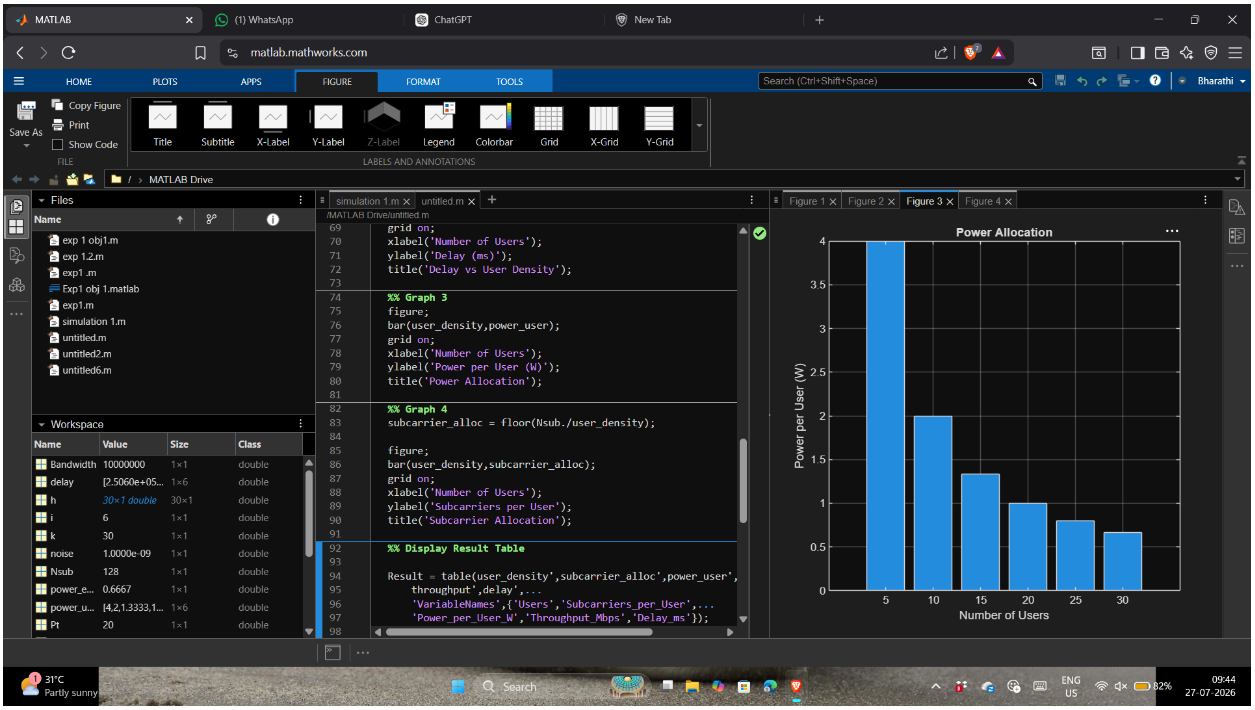
```
clc;
clear;
close all;
%% OFDMA Resource Allocation Simulation
% Parameters
Bandwidth = 10e6;          % 10 MHz
Nsub = 128;                % Number of subcarriers
Pt = 20;                   % Total transmit power (Watts)
user_density = [5 10 15 20 25 30];
throughput = zeros(size(user_density));
delay = zeros(size(user_density));
power_user = zeros(size(user_density));
fprintf('\nOFDMA RESOURCE ALLOCATION SIMULATION\n');
fprintf('--------------------------------------\n');
for i = 1:length(user_density)
  users = user_density(i);
  % Equal Subcarrier Allocation
  sub_per_user = floor(Nsub/users);
  % Equal Power Allocation
  power_each = Pt/users;
  % Random Channel Gain
  h = 0.5 + rand(users,1);
  noise = 1e-9;
  rate = zeros(users,1);
  for k=1:users
     SNR = (power_each*h(k))/noise;
     rate(k) = sub_per_user*log2(1+SNR);
```

```matlab
  end
  throughput(i)=sum(rate)/1e6;      % Mbps
  % Delay Model
  delay(i)=1000/throughput(i);      % ms
  power_user(i)=power_each;
  fprintf('Users = %2d | Throughput = %.2f Mbps | Delay = %.2f ms\n',...
      users,throughput(i),delay(i));
end
%% Graph 1
figure;
plot(user_density,throughput,'-ob','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Throughput (Mbps)');
title('Throughput vs User Density');
%% Graph 2
figure;
plot(user_density,delay,'-sr','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Delay (ms)');
title('Delay vs User Density');
%% Graph 3
figure;
bar(user_density,power_user);
grid on;
xlabel('Number of Users');
ylabel('Power per User (W)');
title('Power Allocation');
%% Graph 4
subcarrier_alloc = floor(Nsub./user_density);
figure;
bar(user_density,subcarrier_alloc);
grid on;
xlabel('Number of Users');
ylabel('Subcarriers per User');
title('Subcarrier Allocation');
%% Display Result Table
Result = table(user_density',subcarrier_alloc',power_user',...
  throughput',delay',...
```

'VariableNames',{'Users','Subcarriers_per_User',...
'Power_per_User_W','Throughput_Mbps','Delay_ms'});
disp(Result);



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Subtitle
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Y-Label
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Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
MATLAB Drive
Files
exp 1 obj1.m
exp 1.2.m
exp1 .m
Exp1 obj 1.matlab
exp1.m
simulation 1.m
untitled.m
untitled2.m
untitled6.m
Workspace
Name Value Size Class
Bandwidth 10000000 1×1 double
delay [2.5060e+05... 1×6 double
h 30×1 double 30×1
i 6 1×1 double
k 30 1×1 double
noise 1.0000e-09 1×1 double
Nsub 128 1×1 double
power_e... 0.6667 1×1 double
power_u... [4,2,1.3333,1... 1×6 double
Pt 20 1×1 double
simulation 1.m
untitled.m
/MATLAB Drive/untitled.m
grid on;
xlabel('Number of Users');
ylabel('Delay (ms)');
title('Delay vs User Density');
%% Graph 3
figure;
bar(user_density,power_user);
grid on;
xlabel('Number of Users');
ylabel('Power per User (W)');
title('Power Allocation');
%% Graph 4
subcarrier_alloc = floor(Nsub./user_density);
figure;
bar(user_density,subcarrier_alloc);
grid on;
xlabel('Number of Users');
ylabel('Subcarriers per User');
title('Subcarrier Allocation');
%% Display Result Table
Result = table(user_density',subcarrier_alloc',power_user',
throughput',delay',...
'VariableNames',{'Users','Subcarriers_per_User',...
'Power_per_User_W','Throughput_Mbps','Delay_ms'});
Figure 1
Figure 2
Figure 3
Figure 4
Power Allocation
Power per User (W)
Number of Users
0 0.5 1 1.5 2 2.5 3 3.5 4
5 10 15 20 25 30
31°C
Partly sunny
Search
ENG
US
82%
09:44
27-07-2026

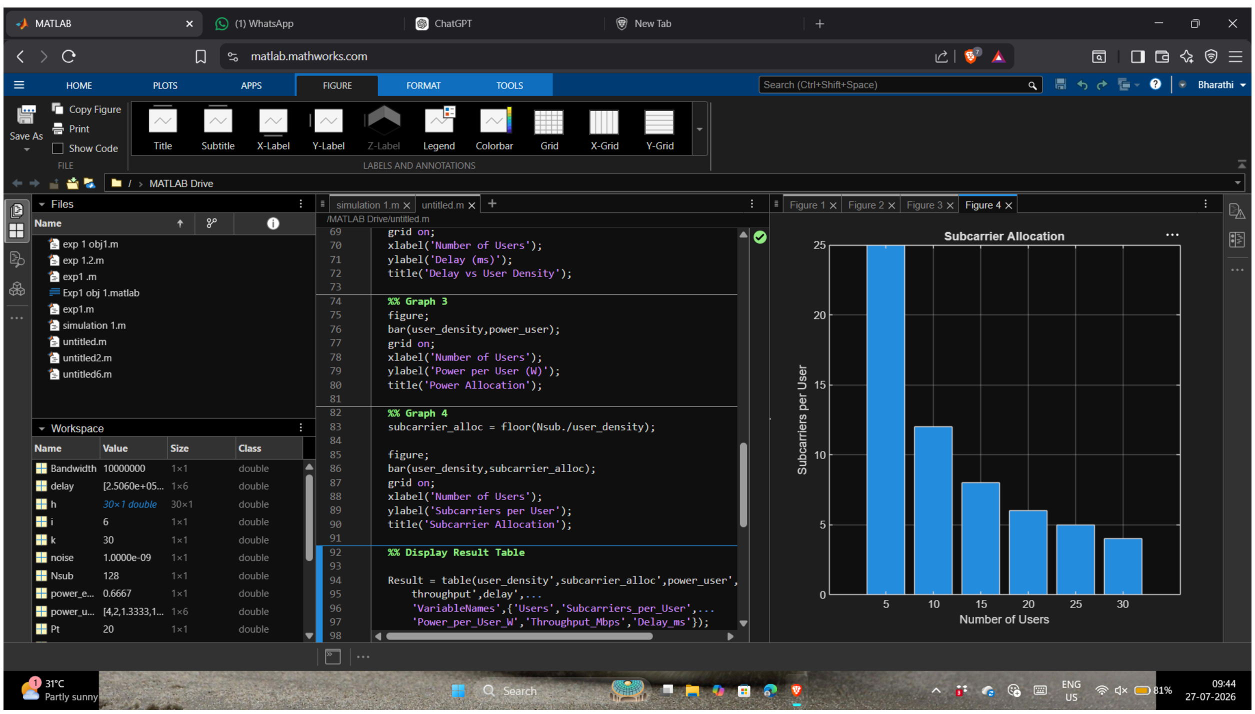
MATLAB
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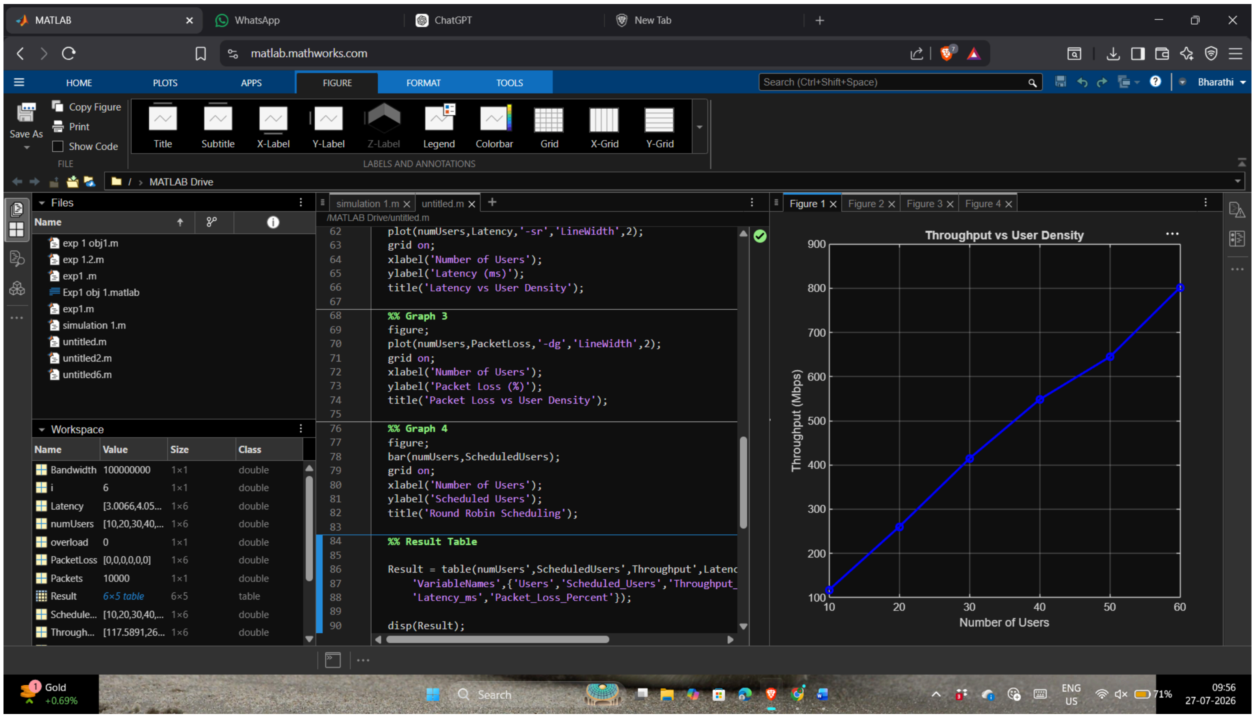
Figure 1
Figure 2
Figure 3
Figure 4
Subcarrier Allocation
Subcarriers per User
Number of Users
0 5 10 15 20 25
5 10 15 20 25 30
31°C
Partly sunny
ENG
US
81%
09:44
27-07-2026

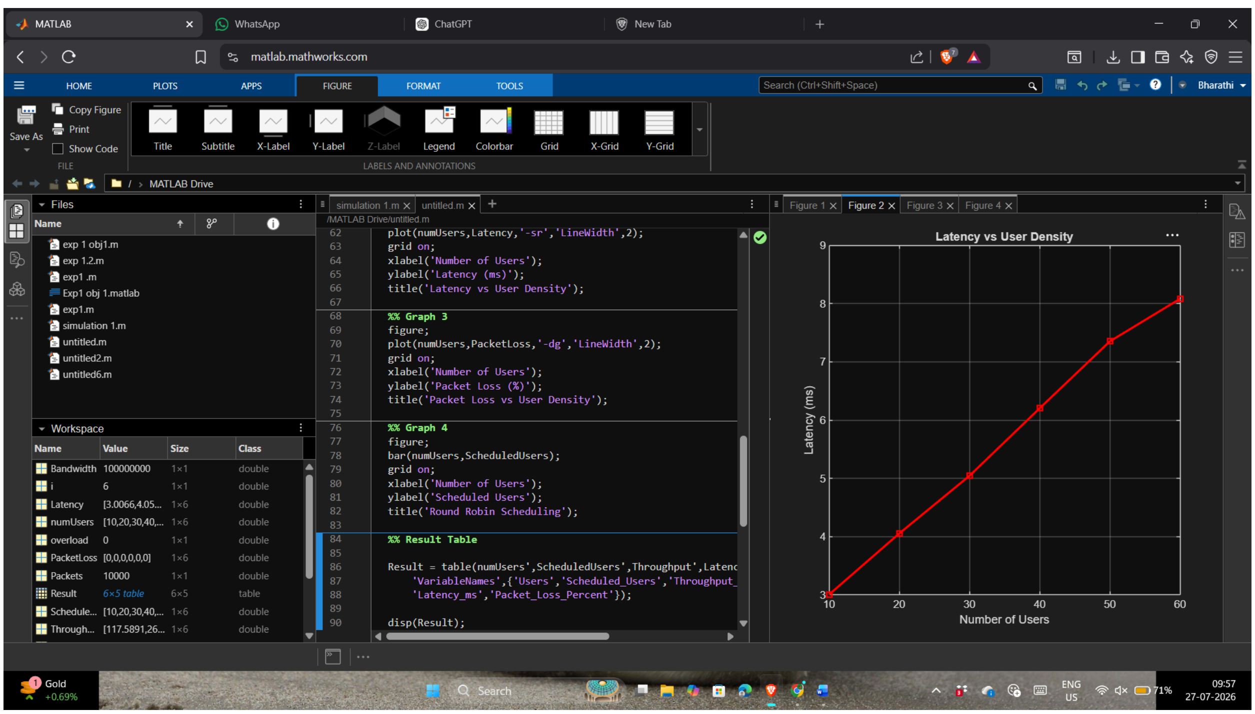
21. Simulation of gNodeB traffic handling under user density, latency, throughput, packet loss, and scheduling

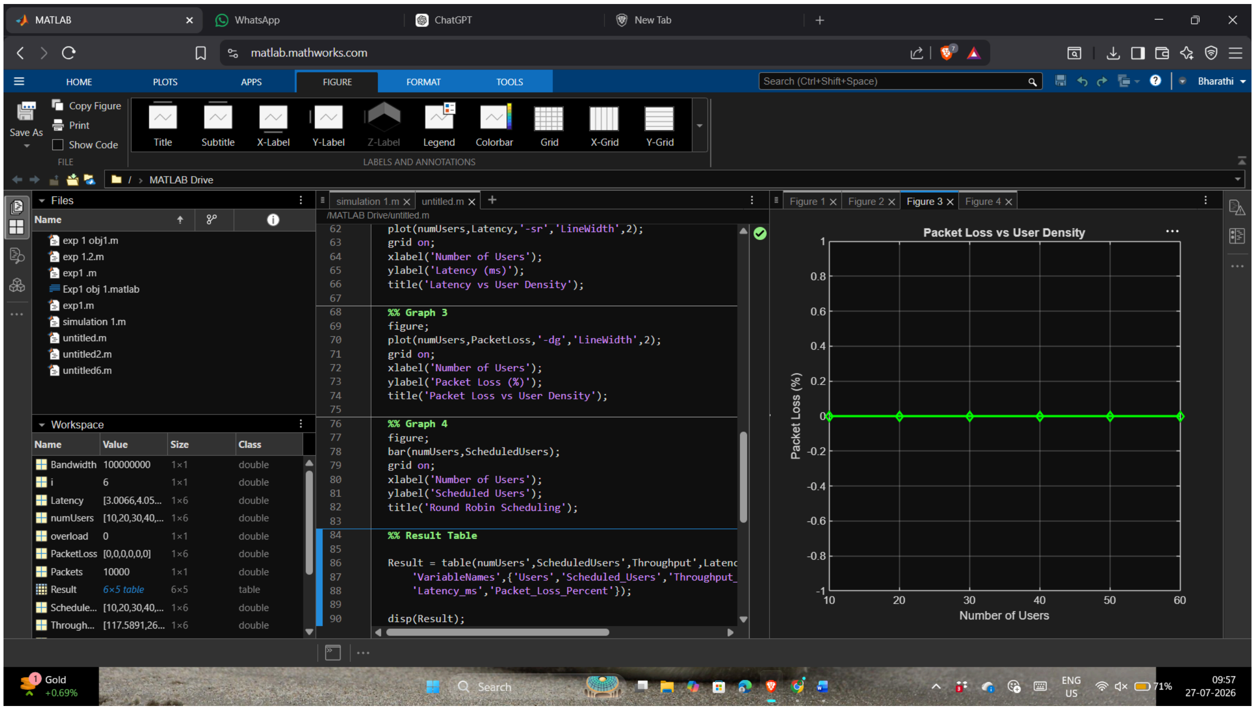
```
clc;
clear;
close all;

%% gNodeB Traffic Handling Simulation

% Parameters
numUsers = [10 20 30 40 50 60];     % User Density
Bandwidth = 100e6;                  % 100 MHz
TotalCapacity = 1000;               % Mbps
Packets = 10000;                    % Total packets

Throughput = zeros(size(numUsers));
Latency = zeros(size(numUsers));
PacketLoss = zeros(size(numUsers));
ScheduledUsers = zeros(size(numUsers));

fprintf('\n5G gNodeB Traffic Handling Simulation\n');
fprintf('----------------------------------------\n');

for i = 1:length(numUsers)

    users = numUsers(i);

    %% Round Robin Scheduling
    ScheduledUsers(i) = users;

    %% Capacity Shared Among Users
    userCapacity = TotalCapacity/users;

    %% Random Traffic Demand
    trafficDemand = 8 + rand(users,1)*12;      % Mbps

    %% Actual Throughput
    throughputUsers = min(userCapacity,trafficDemand);
```

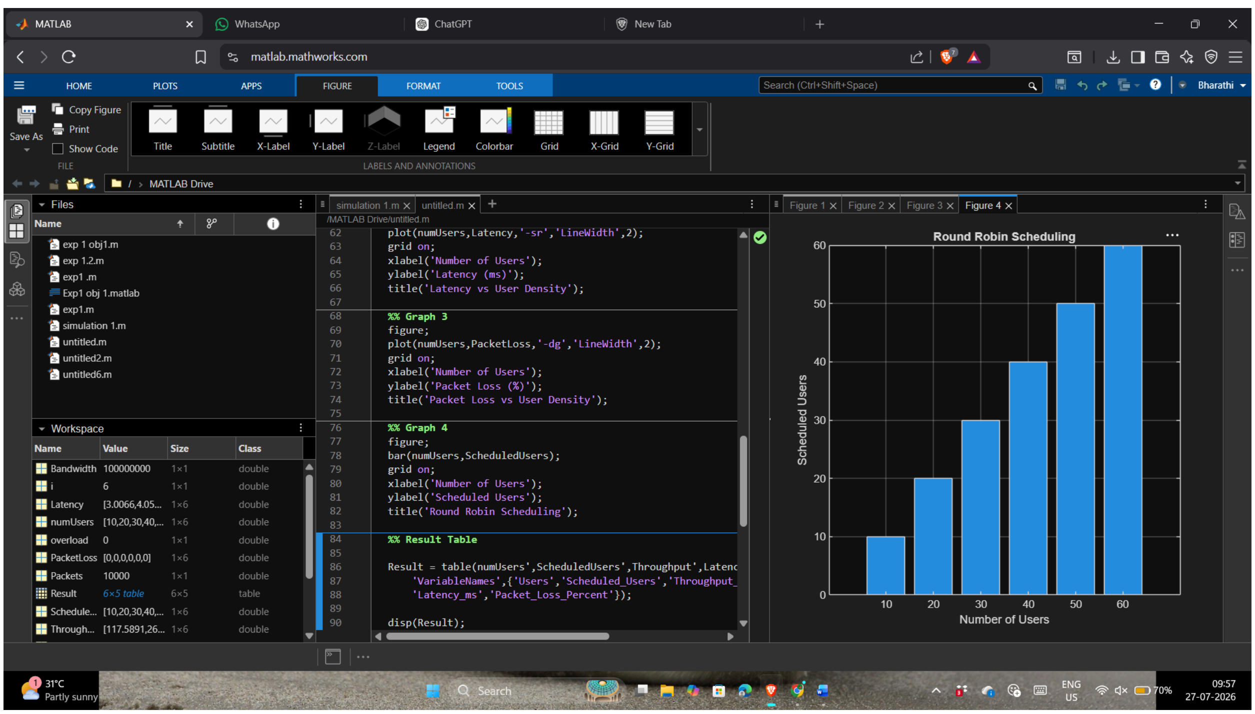
```
    Throughput(i) = sum(throughputUsers);

    %% Latency Model
    Latency(i) = 2 + (users/10) + rand()*0.5;

    %% Packet Loss Model
    overload = max(0,sum(trafficDemand)-TotalCapacity);

    PacketLoss(i) = (overload/TotalCapacity)*100;

    fprintf('Users=%2d  Throughput=%.2f Mbps  Latency=%.2f ms  Packet Loss=%.2f %%\n',...
        users,Throughput(i),Latency(i),PacketLoss(i));

end

%% Graph 1
figure;
plot(numUsers,Throughput,'-ob','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Throughput (Mbps)');
title('Throughput vs User Density');

%% Graph 2
figure;
plot(numUsers,Latency,'-sr','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Latency (ms)');
title('Latency vs User Density');

%% Graph 3
figure;
plot(numUsers,PacketLoss,'-dg','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Packet Loss (%)');
title('Packet Loss vs User Density');

%% Graph 4
```

```
figure;
bar(numUsers,ScheduledUsers);
grid on;
xlabel('Number of Users');
ylabel('Scheduled Users');
title('Round Robin Scheduling');

%% Result Table

Result = table(numUsers',ScheduledUsers',Throughput',Latency',PacketLoss',...
    'VariableNames',{'Users','Scheduled_Users','Throughput_Mbps',...
    'Latency_ms','Packet_Loss_Percent'});

disp(Result);
```


plot(numUsers,Latency,'-sr','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Latency (ms)');
title('Latency vs User Density');
%% Graph 3
figure;
plot(numUsers,PacketLoss,'-dg','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Packet Loss (%)');
title('Packet Loss vs User Density');
%% Graph 4
figure;
bar(numUsers,ScheduledUsers);
grid on;
xlabel('Number of Users');
ylabel('Scheduled Users');
title('Round Robin Scheduling');
%% Result Table
Result = table(numUsers',ScheduledUsers',Throughput',Latenc
'VariableNames',{'Users','Scheduled_Users','Throughput_
'Latency_ms','Packet_Loss_Percent'});
disp(Result);
Throughput vs User Density
Throughput (Mbps)
Number of Users


plot(numUsers,Latency,'-sr','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Latency (ms)');
title('Latency vs User Density');
%% Graph 3
figure;
plot(numUsers,PacketLoss,'-dg','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Packet Loss (%)');
title('Packet Loss vs User Density');
%% Graph 4
figure;
bar(numUsers,ScheduledUsers);
grid on;
xlabel('Number of Users');
ylabel('Scheduled Users');
title('Round Robin Scheduling');
%% Result Table
Result = table(numUsers',ScheduledUsers',Throughput',Latenc
'VariableNames',{'Users','Scheduled_Users','Throughput_
'Latency_ms','Packet_Loss_Percent'});
disp(Result);
Latency vs User Density
Latency (ms)
Number of Users


MATLAB
matlab.mathworks.com
simulation 1.m
untitled.m
Figure 3
Packet Loss vs User Density
Packet Loss (%)
Number of Users
plot(numUsers,Latency,'-sr','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Latency (ms)');
title('Latency vs User Density');
%% Graph 3
figure;
plot(numUsers,PacketLoss,'-dg','LineWidth',2);
grid on;
xlabel('Number of Users');
ylabel('Packet Loss (%)');
title('Packet Loss vs User Density');
%% Graph 4
figure;
bar(numUsers,ScheduledUsers);
grid on;
xlabel('Number of Users');
ylabel('Scheduled Users');
title('Round Robin Scheduling');
%% Result Table
Result = table(numUsers,ScheduledUsers,Throughput',Latenc
'VariableNames',{'Users','Scheduled_Users','Throughput_
'Latency_ms','Packet_Loss_Percent'});
disp(Result);


MATLAB
matlab.mathworks.com
Figure 4
Round Robin Scheduling
Scheduled Users
Number of Users